

Outline Draft

Loveless, Nix, Stokes, Abernathy's

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1 Introduction

Alzheimer's Disease (AD) is characterized as a complex neurodegenerative disease affecting approximately more than 55 million people globally [7]. Symptoms are caused by the degeneration of the central nervous system, and are portrayed by progressive memory loss, difficulty with language, drastic mood swings, among other effects. It is a disease that causes an extensive burden to those suffering from the disease through disorientation, to relatives in the form of mental taxation, as well as economically; projecting an estimated global economic burden of 16.9 trillion dollars by 2050 [9]. These factors are but a few reasons as to the importance of research in this area, where many scientists have worked to understand the underlying mechanisms of AD. However, much is still unknown in how AD is caused, as well as what biological processes occur in the brain, making the disease difficult to come to a decisive conclusion.

Many studies have been conducted in regard to understanding Alzheimer's Disease. With this, there are a few main hypotheses that are generally considered to be factors in the causation and progression of AD [13]. The so-called *Amyloid Cascade Hypothesis* is among the most extensively studied in the literature, where it is proposed that a misfolding of β -amyloid ($A\beta$) protein form senile plaques. These plaques are hypothesized to be the main driving force in the intracellular deposition of misfolded tau proteins in neurofibrillary tangles (NFTs) and memory loss in the progression of Alzheimer's Disease [4]. Growing evidence however, shows that this hypothesis is suggested to be but a small factor in the larger picture of the underlying operative mechanisms of the disease [5]. This has led researchers to branch into other pathways concerning AD, looking at the interactions between $A\beta$ and tau in an attempt to further understand the pathogenesis of these two factors [12]. Some studies have also observed the role of Calcium in this process, further suggesting a more complex process that is occurring biologically. This occurs through a feedback interaction between an upregulation of Ca^{2+} and the amyloidogenic pathway, remodeling the Ca^{2+} signaling pathway [2].

The various complex interactions in AD have shown to be an obstacle to yet overcome when modeling the behavior of the disease. Numerous different studies have in turn given an attempt at creating mathematical models observing different aspects of the disease in order to observe the dynamics in occurrence. This has led to many models being introduced, each aiming to understand various components in Alzheimer's Disease through different methodologies. Compartments concerning the interaction between $A\beta$ and tau are studied, observing the suspected effects on neurodegeneration and cognitive decline through a mechanistic causal model [11, 3]. The feedback loop between $A\beta$ and Ca^{2+} has also been analyzed

using a deterministic system, adding a noise-inducing element to the system [10, 14]. Models that observe lower-level processes have also been studied, lending to valuable insight as to what is suggested to be seen on the cellular level [8, 6].

It should be noted that in these models, while each has its own way of observing some of the underlying mechanisms, there are still potential shortfalls when it comes to finding effective treatments. Evidence suggests that focusing solely on anti-amyloid beta treatments or tau protein treatments, rather attempting to determine treatments for many different aspects of the disease [1, 12] This provides difficulty in understanding exactly the importance of which component of the disease should be the focus of further study in order to provide evidence for treatments having higher efficacy. In this study, a mechanistic model is proposed, analyzing the interaction between $A\beta$, Ca^{2+} , and tau proteins; observing the effects of this multifaceted interaction on neurodegeneration, and cognitive decline. “Sensitivity analysis is performed in order to better understand the importance of each aspect of the model, and optimal control is used in order to determine the suggested avenues of effective treatment.”

2 Equations

Potential systems consist of various mechanisms, but the main differences come from whether or not we contain a Holling-Type III term in our $A\beta$ compartment. The other difference comes from whether or not we include a logistic growth term in each compartment [11], [10]. The Equations including the Holling-Type III term is given by

$$\frac{dA\beta}{dt} = \lambda_{A\beta}A\beta(K_{A\beta} - A\beta) + \lambda_{A\beta Ca}\frac{Ca^2}{Ca^2 + \sigma^2} - (1 - \alpha_{A\beta})\delta_{A\beta}A\beta \quad (1)$$

$$\frac{dCa}{dt} = \lambda_{Ca}Ca(K_{Ca} - Ca) + \lambda_{CaA\beta} - (1 - \alpha_{Ca})\delta_{Ca}Ca \quad (2)$$

$$\frac{d\tau}{dt} = \lambda_{\tau}\tau(K_{\tau} - \tau) + \lambda_{\tau A\beta}A\beta + \lambda_{\tau Ca} - (1 - \alpha_{\tau})\delta_{\tau}\tau \quad (3)$$

$$\frac{dN}{dt} = \lambda_N N(K_N - N) + \lambda_{N\tau}\tau \quad (4)$$

$$\frac{dC}{dt} = \lambda_C C(K_C - C) + \lambda_{CN}NR + \lambda_{C\epsilon}\epsilon. \quad (5)$$

A simpler model can be defined, excluding the Holling-Type III term within $dA\beta/dt$.

$$\frac{dA\beta}{dt} = a_1 - k_1A\beta + \lambda_{A\beta Ca} - (1 - \alpha_{A\beta})\delta_{A\beta}A\beta \quad (6)$$

$$\frac{dCa}{dt} = a_2 - k_2Ca + \lambda_{CaA\beta} - (1 - \alpha_{Ca})\delta_{Ca}Ca \quad (7)$$

$$\frac{d\tau}{dt} = a_3 - k_3\tau + \lambda_{\tau A\beta}A\beta + \lambda_{\tau Ca} - \underbrace{(1 - \alpha_{\tau})\delta_{\tau}\tau}_{\text{off the dome}} \quad (8)$$

$$\frac{dN}{dt} = a_4 - k_4N + \lambda_{N\tau}\tau \quad (9)$$

$$\frac{dC}{dt} = \underbrace{a_5 - k_5C}_{\text{Pal}} + \underbrace{\lambda_{CN}NR + \lambda_{C\epsilon}\epsilon}_{\text{Petrella}}. \quad (10)$$

In each system, a_n, k_n is the growth and wash out, λ_{ab} is the growth rate for a with

respect to b , $ab \neq 0$. α_n is the inefficiency of some drug for n , $0 < \alpha_n < 1$. δ_n is the strength of some drug, $0 < \delta_n < 1$.

The proposed **MODEL TO END ALL MODELS**:

$$\frac{dA_\beta}{dt} = a_1 - r_1 A_\beta + a_2 \frac{Ca^2}{Ca^2 + \sigma^2} \quad (11)$$

$$\frac{dCa}{dt} = b_1 - r_2 Ca + b_2 A_\beta \quad (12)$$

$$\frac{d\tau}{dt} = c_1 - r_3 \tau + c_2 A_\beta + c_3 Ca \quad (13)$$

$$\frac{dN}{dt} = d_1 - k_4 N + d_2 \tau \quad (14)$$

$$\frac{dC}{dt} = e_1 - k_5 C + e_2 NR. \quad (15)$$

where $r_n = k_n + (1 - \alpha_n)\delta_n$.

2.1 Week 3 Equations Currently in work

$$\frac{dA_\beta}{dt} = a_1 - r_1 A_\beta + a_2 \frac{Ca^2}{Ca^2 + \sigma^2} \quad (16)$$

$$\frac{dCa}{dt} = b_1 - r_2 Ca + b_2 A_\beta \quad (17)$$

$$\frac{d\tau}{dt} = c_1 - r_3 \tau + c_2 A_\beta + c_3 Ca \quad (18)$$

$$\frac{dN}{dt} = d_1 \tau \left(1 - \frac{N}{K_N}\right) \quad (19)$$

$$\frac{dC}{dt} = (e_1 N + e_2 \tau) \left(1 - \frac{C}{K_C}\right) \quad (20)$$

and

$$\frac{dA_\beta}{dt} = a_1 - r_1 A_\beta + a_2 \frac{Ca}{Ca + \sigma} \quad (21)$$

$$\frac{dCa}{dt} = b_1 - r_2 Ca + b_2 A_\beta \quad (22)$$

$$\frac{d\tau}{dt} = c_1 - r_3 \tau + c_2 A_\beta + c_3 Ca \quad (23)$$

$$\frac{dN}{dt} = d_1 - k_4 N + d_2 \tau \quad (24)$$

$$\frac{dC}{dt} = e_1 - k_5 C + e_2 NR + e_3 \tau. \quad (25)$$

Var	Value	Unit	Source
A_β	~ 0.01	C	[11]
Ca	0		
τ	0	C	[11]
N	0		
C	0		
a_1	0.25	c/yr	[10]
a_2	2.89	c/yr	[10]
b_1	11	c/yr	[10]
b_2	1000	yr^{-1}	[10]
c_1			
c_2	0.15	yr^{-1}	[3]
c_3			
d_1			
d_2			
e_1	1.67	yr^{-1}	[3]
e_2	3.83	yr^{-1}	[3]
k_1	0.35	yr^{-1}	[10]
k_2	500	yr^{-1}	[10]
k_3			
k_N	1.00	C	
k_C	169.48	C	
α_{A_β}			
α_{Ca}			
α_τ			
δ_{A_β}			
δ_{Ca}			
δ_τ			
σ	0.00027	C	[14]

2.2 Mass Action Terms Model

$$\frac{dA_\beta}{dt} = a_1 - c_1 A_\beta - k_1 A_\beta \tau - \alpha_1 A_\beta \quad (26)$$

$$\frac{dCa}{dt} = b_1 + k_2 A_\beta \tau - c_2 \tau - \alpha_2 \tau \quad (27)$$

$$\frac{d\tau}{dt} = k_3 A_\beta + k_4 \tau - c_3 Ca - \alpha_3 Ca \quad (28)$$

$$\frac{dN}{dt} = rN \left(1 - \frac{N}{K}\right) - k_5 Ca N - k_6 \tau N. \quad (29)$$

3 Parameters

References

- [1] Altaf A Abdulkhaliq et al. “Amyloid- β and Tau in Alzheimer’s disease: pathogenesis, mechanisms, and interplay”. In: *Cell Death & Disease* 17.1 (2026), p. 21.

- [2] Michael J Berridge. “Calcium hypothesis of Alzheimer’s disease”. In: *Pflügers archiv-european journal of physiology* 459.3 (2010), pp. 441–449.
- [3] Wenrui Hao, Suzanne Lenhart, and Jeffrey R Petrella. “Optimal anti-amyloid-beta therapy for Alzheimer’s disease via a personalized mathematical model”. In: *PLoS computational biology* 18.9 (2022), e1010481.
- [4] John A Hardy and Gerald A Higgins. “Alzheimer’s disease: the amyloid cascade hypothesis”. In: *Science* 256.5054 (1992), pp. 184–185.
- [5] Kasper P Kepp et al. “The amyloid cascade hypothesis: an updated critical review”. In: *Brain* 146.10 (2023), pp. 3969–3990.
- [6] Ruoyun Lang, Yuanshun Tan, and Yu Mu. “Stationary distribution and extinction of a stochastic Alzheimer’s disease model”. In: *AIMS Mathematics* 8.10 (2023), pp. 23313–23335.
- [7] Mitali Maji and Subhas Khajanchi. “Mathematical models on Alzheimer’s disease and its treatment: A review”. In: *Physics of Life Reviews* 52 (2025), pp. 207–244.
- [8] Mitali Maji, Laurent Pujo-Menjouet, and Subhas Khajanchi. “Quantifying uncertainty and sensitivity in an Alzheimer’s disease model: A mathematical approach”. In: *Acta Biotheoretica* 74.1 (2026), p. 5.
- [9] Arindam Nandi et al. “Global and regional projections of the economic burden of Alzheimer’s disease and related dementias from 2019 to 2050: a value of statistical life approach”. In: *EClinicalMedicine* 51 (2022).
- [10] Swadesh Pal and Roderick Melnik. “Nonequilibrium landscape of amyloid-beta and calcium ions in application to Alzheimer’s disease”. In: *Physical Review E* 111.1 (2025), p. 014418.
- [11] Jeffrey R Petrella et al. “Computational causal modeling of the dynamic biomarker cascade in Alzheimer’s disease”. In: *Computational and mathematical methods in medicine* 2019.1 (2019), p. 6216530.
- [12] Huiqin Zhang et al. “Interaction between $A\beta$ and tau in the pathogenesis of Alzheimer’s disease”. In: *International journal of biological sciences* 17.9 (2021), p. 2181.
- [13] Jifa Zhang et al. “Recent advances in Alzheimer’s disease: Mechanisms, clinical trials and new drug development strategies”. In: *Signal transduction and targeted therapy* 9.1 (2024), p. 211.
- [14] Yongxin Zhang and Wendi Wang. “Mathematical analysis for stochastic model of Alzheimer’s disease”. In: *Communications in Nonlinear Science and Numerical Simulation* 89 (2020), p. 105347.